

Math 122 Notes on Linear Algebra, 2024

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Part I

Linear Algebra

Chapter 1

A boatload of definitions

We are ready to embark on the final unit of the course, on linear algebra, and of course we're going to start with a boatload of definitions. Let me emphasize at the outset that our main objects of interest will be vector spaces over fields. Rings and modules will certainly appear (and in any future algebra course you take they will likely be the central objects of study), but on first pass you can focus primarily on fields and vector spaces.

1.1 Rings and fields

Definition 1.1.1 A *ring* R is a set with two laws of composition,

$$+ : R \times R \rightarrow R \quad \text{and} \quad \times : R \times R \rightarrow R$$

satisfying the axioms

- $(R, +)$ is an abelian group, with identity denoted $0 \in R$;
- The operation $\times$ is commutative and associative, with identity denoted $1 \in R$; and
- the two operations satisfy the *distributive law*

$$(a + b)c = ac + bc.$$

(Note that we often write ab for $a \times b$, as we just did.)

Finally, if $(R \setminus \{0\}, \times)$ is actually a group (that is, inverses exist) we say that R is a *field*.

Note that as a consequence of the distributive law, we have $a \times 0 = 0$ for all $a \in R$. Note also that if R is a field, then we have the cancellation law

$$ab = ac \text{ and } a \neq 0 \implies b = c$$

and also

$$ab = 0 \implies a = 0 \text{ or } b = 0$$

but neither of these is necessarily true in a ring.

There are many, many variants of this definition. For one thing, in a ring it is usually not assumed that the multiplicative law of composition $\times$ is abelian; if it is, R is called a *commutative ring*. But virtually all the rings we'll deal with here will be commutative, so I just put it in the definition.

1.1.1 Examples

The standard examples are ones you're familiar with: $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ and $\mathbb{Z}/n$. Of these, $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$ are fields and $\mathbb{Z}$ a ring that is not a field. $\mathbb{Z}/n$ is a field when $n = p$ is a prime, but not otherwise: if $n = ab$ is composite, then $ab = 0$ in $\mathbb{Z}/n$, so $\mathbb{Z}/n$ can't be a field; but if $n = p$ is a prime, for any integer a not divisible by p , we can find integers k and l such that

$$kp + la = 1;$$

so $l = a^{-1}$ in $\mathbb{Z}/p$. One bit of notational nonsense: when we want to think of $\mathbb{Z}/p$ as a field, and not just a group, it's usually denoted $\mathbb{F}_p$.

Note that the $\mathbb{F}_p$ are examples of finite fields, but they aren't the only ones; on the homework you'll see, for example, that there exists a field with 4 elements.

1.1.2 Constructions

Say R is a ring. We can form the *polynomial ring*

$$R[x] := \{a_0 + a_1x + a_2x^2 + \cdots + a_nx^n \mid n \in \mathbb{N} \text{ and } a_i \in R\}$$

Here, the rule for addition is the obvious one, and multiplication is defined by setting $x^i x^j = x^{i+j}$ and applying the distributive law. There is also a similarly defined *polynomial ring in n variables* $R[x_1, \dots, x_n]$ for any n .

Now suppose $R = k$ is a field. The polynomial ring $k[x]$ is again a ring, but not a field; we don't have multiplicative inverses. But we can form an associated field: we define

$$k(x) := \left\{ \frac{p(x)}{q(x)} \mid q(x) \neq 0, \text{ with } \frac{p}{q} = \frac{r}{s} \text{ if } ps = qr \right\}.$$

$k(x)$ is called the *field of rational functions* in one variable over k ; there is an analogous definition of the field $k(x_1, \dots, x_n)$ of rational functions in n variables.

Something to think about: if we start with an arbitrary ring R , we can try to make a field out of R by an analogue of the construction above, that is, set

$$K := \left\{ \frac{a}{b} \mid b \neq 0, \text{ with } \frac{a}{b} = \frac{c}{d} \text{ if } ad = bc \right\}$$

For example, if we started with $\mathbb{Z}$, this would produce the field $\mathbb{Q}$, and if we started with $k[x]$ we'd get $k(x)$. But we run into trouble if R contains nonzero elements $a, b \neq 0$ with $ab = 0$.

Another related construction: if R is a ring, we can form the *power series ring* $R[[x]]$ over R , by setting

$$R[[x]] := \{a_0 + a_1x + a_2x^2 + \dots \mid a_i \in R\}$$

This is again a ring: if we're given two power series

$$f(x) = a_0 + a_1x + a_2x^2 + \dots \quad \text{and} \quad g(x) = b_0 + b_1x + b_2x^2 + \dots$$

we can take their product one term at a time, to arrive at

$$fg(x) = a_0b_0 + (a_0b_1 + a_1b_0)x + (a_0b_2 + a_1b_1 + a_2b_0)x^2 + \dots,$$

defining the multiplicative law of composition on $R[[x]]$. Note that we can't just say, "multiply f and g as functions of x ," since these are not functions!

As with polynomials, even if $R = k$ is a field, the ring $k[[x]]$ won't be a field. But there is a related object that is: we define the field of *formal Laurent series* to be

$$k((x)) := \{a_{-n}x^{-n} + a_{-n+1}x^{-n+1} + \dots + a_{-1}x^{-1} + a_0 + a_1x + \dots \mid a_i \in k\}$$

In other words, we are looking at infinite sums of expressions of the form $a_n x^n$, but having only finitely terms with negative exponent; the multiplicative law of composition is defined just as with ordinary power series.

Exercise 1.1.2 Verify that $k((x))$ is indeed a field. What goes wrong if we allow infinitely many terms with negative exponent?

Exercise 1.1.3 Let $k[[x, y]]$ be the ring of power series in two variables over the field k , and consider two subrings

$$k[x][[y]] \quad \text{and} \quad k[[y]][x].$$

Are they the same?

One more example! Suppose you like to work with the field $\mathbb{Q}$ of rational numbers, but are frustrated by the fact that there is no square root of 2 in $\mathbb{Q}$. Well, we can fix that. We construct a new field, denoted $\mathbb{Q}(\sqrt{2})$ and defined to be

$$\mathbb{Q}(\sqrt{2}) := \{a + bx \mid a, b \in \mathbb{Q}\}.$$

The additive law of composition will be termwise; that is,

$$(a + bx) + (c + dx) = (a + c) + (b + d)x$$

and for multiplication we just invoke the distributive law and set $x^2 = 2$; in other words,

$$(a + bx)(c + dx) = (ac + 2bd) + (ad + bc)x.$$

Exercise 1.1.4 Check that this is a field! (It's not that hard, but it's not obvious, either.)

So now we have a number system that includes a square root of 2, with minimal alteration. Note that this would not have worked if the field we started with already contained a square root of 2: for example, if we tried to carry out this construction on a field like $\mathbb{R}$, we would discover that

$$(\sqrt{2} + x)(\sqrt{2} - x) = 0$$

so this couldn't be a field!

1.1.3 Homomorphisms, etc.

Let R and S be rings. We define a *homomorphism* $\phi : R \rightarrow S$ to be a map that respects both laws of composition; that is, for all a and $b \in R$,

$$\phi(a + b) = \phi(a) + \phi(b) \quad \text{and} \quad \phi(ab) = \phi(a)\phi(b).$$

Note that as a consequence, we must have $\phi(0) = 0$ and $\phi(1) = 1$,¹ and it follows that

$$\phi(-a) = -\phi(a) \quad \text{and} \quad \phi\left(\frac{1}{a}\right) = \frac{1}{\phi(a)}$$

One important consequence of this in turn is that *any homomorphism $\phi : K \rightarrow L$ of fields is injective*: for any $a \neq 0 \in K$, if $\phi(a) = 0$ we would have

$$1 = \phi(a \cdot a^{-1}) = \phi(a)\phi(a^{-1}) = 0.$$

1.2 Modules and vector spaces

We'll do the case of vector spaces first, since those are the objects we'll be primarily interested in. To start, fix a field k .

As you know, objects in abstract mathematics tend to be sets with some additional structure, where the structure reflects the sort of operations we carry out on them. With vectors, we primarily do two things: we add them, and we multiply them by scalars. So without further ado here is our definition:

Definition 1.2.1 A *vector space* V over the field k is a set, together with two maps

$$+ : V \times V \rightarrow V \quad \text{and} \quad \times : k \times V \rightarrow V$$

such that $(V, +)$ is an abelian group, and such that for all $a, b \in k$ and $v, w \in V$,

$$\begin{aligned} (a + b)v &= av + bv; \\ a(v + w) &= av + aw; \text{ and} \\ (ab)v &= a(bv). \end{aligned}$$

¹Apologies: here I'm using the same symbols 0 and 1 to denote the identity elements in R and S . It probably won't be the worst notational crime I'll commit.

Next, suppose that V is a vector space over the field k . We say that a subset $W \subset V$ is a *subspace* if it's closed under the two operations above; in other words, if $W \subset V$ is a subgroup with respect to the law of composition given by $+$, and for all $w \in W$ and $\lambda \in k$ we have $\lambda w \in W$.

This generalizes to the notion of a *homomorphism* (a.k.a. linear map) between vector spaces V and W over a field k : we say that a map $\phi : W \rightarrow V$ is linear if it's a homomorphism of abelian groups, and respects scalar multiplication, i.e. for all $w \in W$ and $\lambda \in k$ we have $\phi(\lambda w) = \lambda \phi(w)$. Thus an injective homomorphism $\phi : W \rightarrow V$ is the same thing as a subspace of V .

1.2.1 Basic examples

As always, fix a field k . The most basic example of a vector space over k is

$$k^n := \{(a_1, a_2, \dots, a_n) \mid a_i \in k\}$$

with addition and scalar multiplication given termwise; i.e.,

$$(a_1, \dots, a_n) + (b_1, \dots, b_n) = (a_1 + b_1, \dots, a_n + b_n)$$

and

$$\lambda(a_1, \dots, a_n) = (\lambda a_1, \dots, \lambda a_n).$$

We also have two infinite versions of this: for example, we can define

$$k[x] = \{(a_1, a_2, \dots) \mid a_i \in k \text{ and } a_i = 0 \text{ for all but finitely many } i\}$$

(it's like we took the ring $k[x]$ and forgot how to multiply polynomials, remembering only how to multiply a polynomial by a scalar); and similarly

$$k[[x]] = \{(a_1, a_2, \dots) \mid a_i \in k\}$$

In general, if S is any set, we can give the set

$$k^S := \{\text{maps } S \rightarrow k\}$$

the structure of a vector space over k in the obvious way, setting

$$(f + g)(s) = f(s) + g(s) \quad \text{and} \quad (\lambda f)(s) = \lambda f(s).$$

We can then get subspaces of k^S by imposing suitable restrictions on the maps $f : S \rightarrow k$: for example, if $k = \mathbb{R}$ and $S = [0, 1] \subset \mathbb{R}$ you can verify that the subset of *continuous* maps is a vector space, as is the subset of differentiable functions, etc.

Finally, one more variant of this: for any set S , we can define a vector space

$$k^{[S]} := \{ \text{maps } f : S \rightarrow k \mid f(s) = 0 \text{ for all but finitely many } s \in S \}$$

1.3 Span, independence and bases

Let's start with the notion of the span of a subset $S \subset V$ of a vector space. We can do this abstractly: we define the *span* $\langle S \rangle$ of S to be the smallest subspace of V containing S (or, equivalently, the intersection of all subspaces containing S); we say that S *spans* V if $\langle S \rangle = V$. More concretely, we have a linear map $\phi : k^{[S]} \rightarrow V$ defined by

$$\phi(f) = \sum_{v \in S} f(v) \cdot v;$$

the span of S is simply the image of this map and we say that S spans V if ϕ is surjective.

That in turn suggests two further definitions (and then we'll be done!). First, we say that the subset $S \subset V$ is *independent* if the map ϕ is injective; in concrete terms, this is equivalent to saying that for any $v_1, \dots, v_n \in S$ and $a_1, \dots, a_n \in k$, we have

$$a_1 v_1 + \dots + a_n v_n = 0 \implies a_1 = \dots = a_n = 0.$$

Finally, if ϕ is an isomorphism, then we say that S is a *basis* for V ; concretely, this means that *every element of V is uniquely expressible as a linear combination of elements of S* . We say that V is *finite-dimensional* if there exists a finite spanning set for V .

Exercise 1.3.1 Think of $k[x]$ and $k[[x]]$ as vector spaces over k . Show that the subset $S = \{1, x, x^2, x^3, \dots\}$ is a basis for $k[x]$, but not for $k[[x]]$. Can you find a basis for $k[[x]]$?

This has been a long slog through a lot of definitions. Just to have something resembling a theorem, let me state the first thing we're going to prove on Friday:

Theorem 1.3.2 *If V is finite-dimensional, any two bases for V have the same cardinality.*

The number of elements in a basis for V is called the *dimension* of V . Between now and Friday, think about how you might prove this theorem, and ask yourself: is the statement still true if we drop the hypothesis that V is finite-dimensional?

Chapter 2

Dimension

We want to continue our discussion of vector spaces and linear maps between them. Before we get into that, though, there is one important property of fields we should mention

2.1 The characteristic of a field

Suppose that k is any field, and as usual we denote the multiplicative unit in k by 1. We can ask a simple question: what is the order of 1 in the abelian group $(k, +)$? In other words, we can define a map $\phi : \mathbb{Z} \rightarrow k$ by sending $n \in \mathbb{Z}$ to the sum $n \cdot 1 := 1 + 1 + \cdots + 1$ of 1 with itself n times (or to the sum of -1 with itself $-n$ times, if n is negative). (This is a homomorphism of rings, but that won't matter.) There are two possibilities:

- ϕ is injective; or
- the kernel of ϕ is the subgroup $n\mathbb{Z}$ for some n .

Note that in the second case, the number n must be a prime p : if $n = ab$, by the distributive axiom we have

$$(a \cdot 1)(b \cdot 1) = ab \cdot 1 = 0,$$

so either $a \cdot 1 = 0$ or $b \cdot 1 = 0$.

In the first case above, we say that the field k *has characteristic 0*; in the second case we say that k has characteristic p .

Note that since any homomorphism $\phi : k \rightarrow \ell$ of fields is injective, and takes 1 to 1, we must have $\text{char}(k) = \text{char}(\ell)$. In particular, the characteristic of a field is preserved under all the constructions we discussed last time, like taking the field $k(x)$ of rational functions or the field $k((x))$ of Laurent series over k . In effect, fields of characteristic zero and fields of characteristic p occupy different worlds; many theorems that are true in characteristic 0 are false in characteristic p .

2.2 Dimension

Back to vector spaces! To recall where we were (and maybe introduce some useful terminology), a *vector space* V over a field k is a set with laws

$$+ : V \times V \rightarrow V \quad \text{and} \quad \times : k \times V \rightarrow V$$

such that $(V, +)$ is an abelian group (with identity denoted 0), and satisfying the axioms

$$a(v + w) = av + aw; \quad (a + b)v = av + bv \quad \text{and} \quad a(bv) = (ab)(v)$$

for all $a, b \in k$ and $v, w \in V$.

By way of terminology, if $v_1, \dots, v_n \in V$ are any elements of a vector space V and $a_1, \dots, a_n \in k$ any scalars, we can form the vector

$$v := a_1v_1 + a_2v_2 + \cdots + a_nv_n \in V;$$

this is called a *linear combination* of the vectors $v_1, \dots, v_n$. Thus, for example, we say that a subset $W \subset V$ is a *subspace* if every linear combination of vectors in W is in W . More generally, a map $\phi : V \rightarrow W$ of vector spaces over k is called a *homomorphism*, or a *linear map* (or, if it's clear what's meant, just a *map*), if it preserves linear combinations; that is,

$$\phi(a_1v_1 + a_2v_2 + \cdots + a_nv_n) = a_1\phi(v_1) + \cdots + a_n\phi(v_n)$$

for all $a_i \in k$ and $v_i \in V$.

Let me emphasize one point: *linear combinations are necessarily finite*; the axioms of vector spaces do not give us a way of adding up an infinite collection of vectors (unless of course all but finitely many are 0).

If $\phi : V \rightarrow W$ is any linear map, the *kernel* $\phi^{-1}(0)$ is a subspace of V , and the image of ϕ is a subspace of W .

2.2.1 Bases, span and linear independence

Let's introduce these notions in the opposite order from last time. First of all, if V is a vector space and $\{v_i \in V\}$ any collection of vectors in V , we say that $\{v_i\}$ is a *basis* for V if *every element of V is uniquely expressible as a linear combination of the vectors v_i* . Thus, if $\{v_i\}$ is finite, with n elements, we get an isomorphism $\phi : k^n \longrightarrow V$ defined by

$$\phi : (a_1, \dots, a_n) \mapsto a_1 v_1 + \dots + a_n v_n.$$

For example, in the vector space k^n itself, the elements $e_i = (0, \dots, 0, 1, 0, \dots, 0)$ —defined to be the n -tuple of elements of k with a 1 in the i th place and all others 0—form a basis, called the *standard basis* for k^n .

In general, we say a vector space is *finite-dimensional* if it has a finite basis; that is, by the above, if it is isomorphic to k^n for some n .

In the same terms, we can define two weaker conditions on a subset $S = \{v_i\} \subset V$ of a vector space V . First, we say that S *spans* V if every element of V is expressible as a linear combination of the v_i ; that is, if the map ϕ above is surjective; and we say that the vectors v_i are *linearly independent* if no nontrivial linear combination of them is 0; equivalently, if the map ϕ is injective.

Here are a couple useful observations.

Proposition 2.2.1 *If V is any vector space over the field k and $S = \{v_1, \dots, v_n\} \subset V$ is any finite spanning set, then S contains a basis for V .*

Proof. Since S spans V , to say that it's not already a basis means that the vectors v_i are not linearly independent; that is, there exists a linear relation

$$a_1 v_1 + \dots + a_n v_n = 0$$

with not all $a_i = 0$. Now choose any i for which $a_i \neq 0$; we can solve this equation and write

$$v_i = \frac{-\sum_{j \neq i} a_j v_j}{a_i}$$

so v_i is already in the span of $S' := \{v_j \mid j \neq i\}$; since we assumed that S spanned V , it follows that S' spans V as well. We can continue in this way, getting smaller and smaller subsets that still span V ; since we assumed that S was finite, this process has to stop and when it does we have a basis.

Exercise 2.2.2 Prove the dual statement: that if $S \subset V$ is a linearly independent collection of vectors in a finite-dimensional vector space V , then we can enlarge S to a basis.

Now we can prove the fundamental

Theorem 2.2.3 *Any two bases of a finite-dimensional vector space V have the same cardinality.*

Proof. Suppose that $\{s_1, \dots, s_n\}$ and $\{t_1, \dots, t_m\}$ are both bases for V . Our basic claim is that, for some j , the collection of vectors $\{t_j, s_1, \dots, s_{n-1}\}$ is again a basis—in other words, we can swap out the basis vector s_n for one of the t_i . Carrying this out n times, we arrive at a basis of the same cardinality n consisting entirely of vectors t_i ; since a proper subset of a basis cannot be a basis, this must be the whole collection $\{t_i\}$ and hence $m = n$.

To prove the basic claim, we use an argument like the one above for Proposition 2.2.1. To begin with, we know that the subset $\{s_1, \dots, s_{n-1}\}$ doesn't span, so there must be some j such that t_j is not a linear combination of $s_1, \dots, s_{n-1}$. Since t_j is in the span of $s_1, \dots, s_n$, we can write

$$t_j = a_1 s_1 + \dots + a_n s_n$$

with $a_n \neq 0$; we can then solve this for s_n and write

$$s_n = \frac{t_j - a_1 s_1 - \dots - a_{n-1} s_{n-1}}{a_n}$$

so s_n —and hence all of V —is in the span of $\{t_j, s_1, \dots, s_{n-1}\}$.

Similarly, if the vectors $\{t_j, s_1, \dots, s_{n-1}\}$ failed to be linearly independent, so that there was a nontrivial linear combination

$$ct_j + a_1s_1 + \dots a_{n-1}s_{n-1} = 0$$

the coefficient c would have to be nonzero (since $s_1, \dots, s_{n-1}$ are linearly independent) and so t_j would be in the linear span of $s_1, \dots, s_{n-1}$, contrary to hypothesis.

We can thus define the *dimension* of a finite-dimensional vector space to be the cardinality of any basis. Note that we have already established that any finite-dimensional vector space is isomorphic to k^n for some n , so in effect the content of Theorem 2.2.3 is simply that *for $m \neq n$, the vector space k^n is not isomorphic to k^m .*

One last note before we move on. We'll be dealing almost exclusively with finite-dimensional vector spaces, so we won't get into it, but we should say that infinite-dimensional vector spaces present additional challenges. For example, consider the vector spaces $k[x]$ and $k[[x]]$ —that is,

$$\{(a_0, a_1, \dots) \mid a_i \in k \text{ and } a_i = 0 \text{ for all but finitely many } i\}$$

and

$$\{(a_0, a_1, \dots) \mid a_i \in k\}.$$

There is a natural basis for $k[x]$, namely $\{1, x, x^2, \dots\}$. But it's not clear that a basis for $k[[x]]$ even exists—see if you can find one!

2.3 Products and Homs

We have two main constructions to introduce. The first is a pretty straightforward analogue of a construction we made with groups, the second more interesting.

2.3.1 Products

Let V and W be vector spaces over a field k . We define the *product* $V \times W$ exactly as we did for groups: as a set, we take

$$V \times W := \{(v, w) \mid v \in V \text{ and } w \in W\}.$$

Our two operations are defined in the natural way: we set

$$(v, w) + (v', w') := (v + v', w + w') \quad \text{and} \quad \lambda \cdot (v, w) := (\lambda v, \lambda w).$$

If V and W are finite-dimensional, then we have

$$\dim(V \times W) = \dim(V) + \dim(W);$$

if $\{s_1, \dots, s_n\}$ is a basis for V and $\{t_1, \dots, t_m\}$ is a basis for W , then

$$(s_1, 0), \dots, (s_n, 0), (0, t_1), \dots, (0, t_m)$$

will be a basis for $V \times W$.

Here is one situation in which this arises. Suppose that U is any vector space, and $V, W \subset U$ two subspaces such that

$$V \cup W \text{ spans } U, \quad \text{and} \quad V \cap W = \{0\}.$$

In this setting, U is isomorphic to $V \times W$: our two hypotheses say that the natural map $V \times W \rightarrow U$ sending (u, w) to $u + w$ is surjective and injective, respectively. In this circumstance, we sometimes write $U = V \oplus W$ (though most people use the symbols $V \times W$ and $V \oplus W$ interchangeably).

2.3.2 Homs

As always, let V and W be vector spaces over a field k . We can form a new vector space $\text{Hom}(V, W)$ as follows: as a set,

$$\text{Hom}(V, W) := \{\text{linear maps } \phi : V \rightarrow W\};$$

and the operations of addition and scalar multiplication are defined by

$$(\phi + \psi)(v) := \phi(v) + \psi(v) \quad \text{and} \quad (\lambda\phi)(v) := \lambda\phi(v).$$

So, the set of linear maps $V \rightarrow W$ is itself a vector space. The natural question to ask is, what's its dimension? We'll answer that now by introducing bases $v_1, \dots, v_n$ for V and $w_1, \dots, w_m$ for W . The first thing to say is

that *any* map $\phi : V \rightarrow W$ is determined by the vectors $\phi(v_1), \dots, \phi(v_n)$, and these can be any n vectors in W . In other words, if we write

$$\begin{aligned}\phi(v_1) &= a_{1,1}w_1 + \cdots + a_{m,1}w_m \\ \phi(v_2) &= a_{1,2}w_1 + \cdots + a_{m,2}w_m \\ &\dots \\ \phi(v_n) &= a_{1,n}w_1 + \cdots + a_{m,n}w_m\end{aligned}$$

then every collection of scalars $a_{i,j} \in k$ determines a unique linear map. This gives us an isomorphism of vector spaces

$$\text{Hom}(V, W) \cong k^{mn},$$

and so the dimension of $\text{Hom}(V, W)$ is $\dim(V) \cdot \dim(W)$.

Another way to say this is: for each i and j with $1 \leq i \leq m$ and $1 \leq j \leq n$, define a linear map $\phi_{i,j} : V \rightarrow W$ by

$$\phi_{i,j}(e_j) = w_i \quad \text{and} \quad \phi_{i,j}(e_k) = 0 \text{ for all } k \neq j.$$

What we're saying is that *the linear maps $\phi_{i,j}$ form a basis for $\text{Hom}(V, W)$* , and again we see that the dimension of $\text{Hom}(V, W)$ is mn .

Chapter 3

Linear maps

Before we launch into the main subject matter of today's lecture, let me just mention a basic fact about linear maps and dimensions. First, we recall a statement from last time: if V is a finite-dimensional vector space, then

- Any linearly independent subset of V can be enlarged to a basis; and
- Any spanning set $S \subset V$ contains a basis.

With this said, we have the

Proposition 3.0.1 *If $\phi : V \rightarrow W$ is a linear map between finite-dimensional vector spaces, then*

$$\dim(\ker \phi) + \dim(\operatorname{Im} \phi) = \dim V.$$

Proof. Start by choosing a basis $v_1, \dots, v_k$ for the kernel of ϕ . We can then enlarge $\{v_1, \dots, v_k\}$ to a basis $\{v_1, \dots, v_n\}$ for all of V ; and then we claim that *the vectors $\phi(v_{k+1}), \dots, \phi(v_n)$ form a basis for $\operatorname{Im}(\phi)$.*

To see this, we have to prove that they are independent, and span $\operatorname{Im}(\phi)$. To see that they're independent, suppose that for some scalars $b_{k+1}, \dots, b_n$ we have $b_{k+1}\phi(v_{k+1}) + \dots + b_n\phi(v_n) = 0$. The linear combination $b_{k+1}v_{k+1} + \dots + b_nv_n$ is then in the kernel of ϕ , and so is expressible as a linear combination $b_1v_1 + \dots + b_kv_k$ of $v_1, \dots, v_k$, giving a linear relation among $v_1, \dots, v_n$; since these are independent, the coefficients b_i must all be zero. To see that they span is even easier: any vector $w \in \operatorname{Im} \phi$ is by definition the image of a

vector $b_1v_1 + \cdots + b_nv_n \in V$; we can just drop the first k to see that $w = b_{k+1}\phi(v_{k+1}) + \cdots + b_n\phi(v_n)$.

We do have one more bit of language, but it's a simple one:

Definition 3.0.2 If $\phi : V \rightarrow W$ is a linear map, the *rank* of ϕ is defined to be the dimension of the image $\text{Im}(\phi)$.

3.1 $\text{Hom}(V, W) \cong k^{mn}$

Last time, we introduced a new construction: given two vector spaces V and W over a field k , say of dimensions n and m , we saw that the set of linear maps $\phi : V \rightarrow W$ could be given the structure of a vector space itself, called $\text{Hom}(V, W)$; we also saw, by choosing bases for V and W , that $\text{Hom}(V, W)$ was isomorphic to the vector space k^{mn} of $n \times m$ matrices with entries in k . I want to revisit that, and ask a couple follow-up questions:

- How does the isomorphism $\text{Hom}(V, W) \cong k^{mn}$ depend on the choice of bases? and
- Is there a “normal form” for the matrix representing a given linear map $\phi : V \rightarrow W$, that is, a way of choosing bases for V and W so that the matrix corresponding to ϕ has particularly simple form?

Happily, we'll be able to answer both questions explicitly.

To start, recall how we arrived at the isomorphism $\text{Hom}(V, W) \cong k^{mn}$. Having chosen bases $v_1, \dots, v_n$ for V and $w_1, \dots, w_m$ for W , we simply wrote the images $\phi(v_i)$ of the basis vectors v_i as a linear combination of the w_j :

$$\begin{aligned}\phi(v_1) &= a_{1,1}w_1 + \cdots + a_{m,1}w_m \\ \phi(v_2) &= a_{1,2}w_1 + \cdots + a_{m,2}w_m \\ &\vdots \\ \phi(v_n) &= a_{1,n}w_1 + \cdots + a_{m,n}w_m.\end{aligned}$$

The $a_{i,j}$ determine ϕ : if $v = c_1v_1 + \cdots + c_nv_n \in V$ is any vector in V , by linearity of ϕ we must have

$$\phi(v) = \left(\sum_{k=1}^n c_k a_{1,k}\right)w_1 + \cdots + \left(\sum_{k=1}^n c_k a_{m,k}\right)w_m$$

This gives us the isomorphism $\text{Hom}(V, W) \cong k^{mn}$.

Another way to say this is to start with the observation that if V is a vector space of dimension n , *a basis for V is the same thing as an isomorphism $k^n \cong V$* : given a basis $v_1, \dots, v_n$ for V , we simply send $(c_1, \dots, c_n) \in k^n$ to the corresponding linear combination $\sum c_i v_i$ of the v_i . Given this, if we choose bases for V and W and consider the diagram

$$\begin{array}{ccc} V & \xrightarrow{\phi} & W \\ \cong \downarrow & & \downarrow \cong \\ k^n & \xrightarrow{A} & k^m \end{array}$$

what we see is that *for every $\phi : V \rightarrow W$ there is a unique map $A : k^n \rightarrow k^m$ that makes this commute*; this gives us the isomorphism.

3.1.1 COB matrices

Let's now ask the first of our questions: what happens if we choose different bases for V and W ? Given the diagram above, the answer will probably not be a surprise.

To start, suppose we choose a different basis $v'_1, \dots, v'_n$ for V . To begin with, we can write each of the new basis vectors as a linear combination of the old:

$$\begin{aligned} v'_1 &= p_{1,1}v_1 + \dots + p_{n,1}v_n \\ v'_2 &= p_{1,2}v_1 + \dots + p_{n,2}v_n \\ &\dots \\ v'_n &= p_{1,n}v_1 + \dots + p_{n,n}v_n. \end{aligned}$$

We arrive at an $n \times n$ matrix P , called the *change of basis matrix*.

Exercise 3.1.1 Sanity check: if R is the $n \times n$ matrix expressing the old basis vectors v_i in terms of the new ones v'_j , show that the product PR is the identity matrix.

Now, if A is the matrix corresponding to the map ϕ in terms of our original bases, as above, and we want to find the matrix A' corresponding to ϕ in terms of the new basis for V , we have to say where each of the new basis vectors v'_i goes; we start with the expression

$$v'_1 = p_{1,1}v_1 + \cdots + p_{n,1}v_n;$$

applying ϕ , we get

$$\begin{aligned} \phi(v'_i) &= \phi\left(\sum_{k=1}^n p_{k,i}v_k\right) \\ &= \sum_{k=1}^n p_{k,i}\phi(v_k) \\ &= \sum_{k=1}^n \sum_{j=1}^m p_{k,i}a_{j,k}w_j. \end{aligned}$$

In other words, if A is the matrix associated to ϕ in the basis $\{v_i\}$ and A' is the matrix associated to ϕ in the basis $\{v'_i\}$, then

$$A' = AP$$

where P is the change of basis matrix.

At this point, I've maxed out my ability to deal with matrices (my tolerance is fairly low). So let me just add that if in addition we introduce a new basis $w'_1, \dots, w'_m$ for W , the matrix representative A' of ϕ in terms of both new bases is

$$A' = Q^{-1}AP$$

where Q is the change of basis matrix expressing the new basis vectors w'_i in terms of the w_i (equivalently, Q^{-1} is the change of basis matrix expressing the old basis vectors w_i in terms of the new ones w'_i), and leave it to you to work it out.

3.1.2 Normal form

We now take up the second question above: given a linear map $\phi : V \rightarrow W$, is there a way of choosing bases for V and W in terms of which the matrix representative of ϕ is particularly simple? Thankfully, the answer is yes, and matrix multiplication will not be involved.

To set up, suppose V and W are vector spaces of dimensions n and m and $\phi : V \rightarrow W$ any linear map; we let r be the rank of ϕ , so in particular the kernel of ϕ has dimension $n - r$. We start by choosing a basis for $\ker(\phi)$, which we'll denote $v_{r+1}, \dots, v_n$, and then enlarging this to a basis $v_1, \dots, v_n$ for all of V . As we saw in the proof of Proposition 3.0.1, the images $w_1 := \phi(v_1), \dots, w_r := \phi(v_r)$ then form a basis for the image of ϕ , and we can expand this to a basis $w_1, \dots, w_m$ for W .

In terms of these bases, the map ϕ is very simple: we have

$$\phi(v_i) = \begin{cases} w_i, & \text{if } i \leq r; \text{ and} \\ 0, & \text{if } i > r. \end{cases}$$

Equivalently, the matrix of the map ϕ with respect to these two bases is given in block form as

$$\begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$$

where I_r represents the identity $r \times r$ matrix.

3.2 Quotient spaces

To start, suppose V is a vector space over our field k and $U \subset V$ a subspace. U is in particular a subgroup of the abelian group V , and so we can form the quotient group V/U ; by definition, this is the set of cosets $v + U \subset V$ of U in V . A priori this is just an abelian group, but it's pretty clear how to define a scalar multiplication: we have to set $\lambda \cdot (v + U) := \lambda v + U$. With this, V/U has the structure of a vector space, called the *quotient space* of $U \subset V$.

Note that we have an exact sequence of vector spaces

$$0 \rightarrow U \rightarrow V \rightarrow V/U \rightarrow 0$$

and in particular we see that $\dim(V/U) = \dim V - \dim U$.

One more bit of language: if $\phi : V \rightarrow W$ is any linear map, we define the *cokernel* of ϕ by

$$\text{coker}(\phi) := W/\text{Im}(\phi).$$

We have thus an exact sequence

$$0 \rightarrow \ker(\phi) \rightarrow V \xrightarrow{\phi} W \rightarrow \operatorname{coker}(\phi) \rightarrow 0.$$

One way to think of this: the kernel of a linear map ϕ measures the failure of ϕ to be injective; the cokernel measures its failure to be surjective. We'll see other ways in which the kernel and cokernel of a map are dual constructions in the next section.

3.3 Dual spaces

We start with the definition of the dual of a vector space. This is actually a special case of a construction we've already made: if V is any vector space over k , we define the dual vector space V^* to be

$$V^* := \operatorname{Hom}(V, k) = \{\text{linear maps } \ell : V \rightarrow k\}.$$

Note that if V has basis $v_1, \dots, v_n$, we can define the *dual basis* $\ell_1, \dots, \ell_n$ for V^* by

$$\ell_i(v_j) := \begin{cases} 1, & \text{if } j = i; \text{ and} \\ 0, & \text{if } j \neq i. \end{cases}$$

In particular, if V is finite-dimensional then the dual space V^* is isomorphic to V , but the isomorphism is not canonical; it depends on the choice of a basis for V .

3.3.1 Double duals

To try to make sense of the last sentence, let me contrast it with the case of the double dual. If V is any vector space, the *double dual* of V is what the name suggests: it's the dual of V^* , or in other words the space of linear maps $V^* \rightarrow k$. The key thing to observe is that *an element of V determines a map $V^* \rightarrow k$ by evaluation*: if $v \in V$ is any vector, we can associate to it the map $V^* \rightarrow k$ that sends a linear function $\ell : V \rightarrow k$ its value $\ell(v)$ at v . This gives us a natural map

$$\psi : V \longrightarrow \operatorname{Hom}(V^*, k) = (V^*)^*$$

which in case V is finite-dimensional is an isomorphism. Note that this map does not depend on any choices, or any extra structure (like a basis) that

we might put on V ; that's what we mean by calling the map ψ *natural*, or *canonical*.

3.3.2 The infinite-dimensional case

Note that in the case of V infinite-dimensional, the dual V^* will not necessarily be isomorphic to V . The standard example of this is to take

$$V = k[x] = \{(a_0, a_1, a_2, \dots) \mid a_i \in k \text{ and all but finitely many } a_i = 0\}$$

Here we have a countable basis for V , consisting of the vectors e_i where e_i is the sequence with a 1 in the i th place and zero otherwise; and we correspondingly have linear functions $\ell_i : V \rightarrow k$ where $\ell_i(a) := a_i$. But these don't come close to spanning V^* ; their span doesn't include, for example, the linear function $\ell : V \rightarrow k$ defined by $\ell(a) := \sum a_i$.

In fact, the dual of V in this case is the vector space

$$V^* = k[[x]] = \{(a_0, a_1, a_2, \dots) \mid a_i \in k\}$$

which is much larger than V ; for example, it doesn't have a countable basis.

(To prove that the dual of $V = k[x]$ is $k[[x]]$, note that a linear map $\ell : V \rightarrow k$ is determined by its values on the e_i , and that these values can be an arbitrary sequence of elements of k .)

In the infinite-dimensional case we still have a natural map $V \rightarrow (V^*)^*$, defined as before by sending $v \in V$ to the linear function $V^* \rightarrow k$ sending $\ell \in V^*$ to $\ell(v)$; and this map is in general injective but far from surjective.

3.4 Constructions arising from the dual

3.4.1 Annihilators

This is pretty simple. Suppose V is any vector space, and $U \subset V$ a subspace. The *annihilator* of U is the subspace of the dual space V^* defined by

$$\text{Ann}(U) := \{\ell \in V^* \mid \ell(U) = 0\};$$

in other words, it's the subspace of V^* consisting of linear maps $V \rightarrow k$ whose restriction to U is zero. This is sometimes denoted $U^\perp$, but I would like to reserve that for another construction coming up.

Note that a linear map $\ell : V \rightarrow k$ that is 0 on U is the same thing as a linear map $V/U \rightarrow k$ on the quotient, so we could also write

$$\text{Ann}(U) = (V/U)^*;$$

in particular, we see that $\dim(\text{Ann}(U)) = \dim V - \dim U$.

3.4.2 Transposes

We come now to a fundamental and ubiquitous construction: the *transpose* of a linear map $\phi : V \rightarrow W$. This is a map $\phi^* : W^* \rightarrow V^*$ between the duals of V and W , and going the other way. It's confusing, but easy enough to define: given an element of W^* —that is, a linear map $\ell : W \rightarrow k$ —we send it to the linear map $\ell \circ \phi : V \rightarrow k$ obtained by composing with ϕ . In diagram form,

$$\begin{array}{ccc} V & & \\ \phi \downarrow & \searrow \phi^*(\ell) & \\ W & \xrightarrow{\ell} & k \end{array}$$

What does this have to do with the ordinary notion of the transpose of a matrix? This is simple to say, if messy to verify: if we choose bases v_i and w_j for V and W , then in terms of these bases we can represent the map ϕ by a matrix M . We can also represent the transpose map $\phi^* : W^* \rightarrow V^*$ in terms of the dual bases v_i^* and w_j^* by a matrix N ; and then the matrices M and N will be the transposes of each other.

There is one basic fact we should mention. For finite-dimensional V and W , the construction of the transpose of a map gives an isomorphism

$$\text{Hom}(V, W) \xrightarrow{t} \text{Hom}(W^*, V^*).$$

Given that $(V^*)^* = V$ and $(W^*)^* = W$, we could apply it again to get a map

$$\text{Hom}(W^*, V^*) \xrightarrow{t} \text{Hom}(V, W).$$

What happens if we compose these? We get a natural isomorphism $\text{Hom}(V, W) \rightarrow \text{Hom}(V, W)$, and it won't be too much of a surprise that this is the identity; that is, for any linear map $\phi : V \rightarrow W$ we have

$$(\phi^*)^* = \phi.$$

Exercise 3.4.1 Let $\phi : V \rightarrow W$ be any linear map, and $\phi^* : W^* \rightarrow V^*$ its transpose. Verify that

$$\ker \phi = \text{Ann}(\text{Im} \phi^*) \quad \text{and} \quad \text{Im}(\phi^*) = \text{Ann}(\ker \phi)$$

Chapter 4

Operators

4.1 Preliminaries

4.1.1 Polynomials

Let k be any field. As always, a *polynomial* $f \in k[x]$ is an expression of the form $a_n x^n + \cdots + a_1 x + a_0$; the largest n for which $a_n \neq 0$ is called the *degree* of f . We want to establish two key facts about polynomials, one elementary and one deeper; for the latter, we'll have to defer the proof to another course.

The elementary fact is the

Proposition 4.1.1 *Let $f \in k[x]$ be a polynomial of degree n . If $\lambda \in k$ is any scalar such that $f(\lambda) = 0$, then f is divisible by the polynomial $x - \lambda$; that is, we have*

$$f = (x - \lambda) \cdot g$$

for some polynomial $g \in k[x]$ of degree $n - 1$.

Proof. For any λ , we can carry out long division of polynomials, dividing $(x - \lambda)$ into f to arrive at an expression

$$f = (x - \lambda) \cdot g + c$$

where $c \in k$ is a constant. If $f(\lambda) = 0$, we can plug this into the above equality to deduce that $c = 0$.

A scalar $\lambda \in k$ such that $f(\lambda) = 0$ is called a *root* of the polynomial f . Note that as an immediate consequence of Proposition 4.1.1 we see that *a polynomial f of degree n cannot have more than n roots.*

Note also that a polynomial f defines a function $k \rightarrow k$, but is not necessarily determined by that function. For example, if $k = \mathbb{F}_p$, the polynomial $x^p - x$ has the value $f(\lambda) = 0$ for all $\lambda \in \mathbb{F}_p$, but is not the zero polynomial. Over infinite fields, though, a polynomial *is* determined by its values; this follows from Proposition 4.1.1.

Our second result on polynomials is what's called the Fundamental Theorem of Algebra. It's very simple to state:

Theorem 4.1.2 *Over $k = \mathbb{C}$, every polynomial of positive degree has a root.*

Note that as a consequence, every polynomial $f \in \mathbb{C}[x]$ *factors completely*; that is, we can write

$$f = c \cdot (x - \lambda_1)(x - \lambda_2) \cdots (x - \lambda_n)$$

We get this by applying Theorem 4.1.2 to f to arrive at an expression $f = (x - \lambda)g$, then applying Theorem 4.1.2 to g , and so on.

In general, a field k is said to be *algebraically closed* if it has the property that every polynomial of positive degree over k has a root. You may not know many algebraically closed fields other than $\mathbb{C}$, but they are ubiquitous: one of the theorems you'll see in Math 123 is that *every field is contained in an algebraically closed field*. In fact, you've already seen how to do this: the construction of the field $\mathbb{Q}(\sqrt{2})$ that we gave in Lecture 7 can be adapted in general to construct a field in which any given polynomial $f \in k[x]$ has a root, and the rest is (nontrivial) bookkeeping.

4.1.2 Direct sum decompositions

This is more a matter of introducing useful language than new content. The situation is this: suppose V is any vector space, and $U_1, \dots, U_k \subset V$ any collection of subspaces. We say that the subspaces $\{U_i\}$ *span* V if every element $v \in V$ can be expressed as a sum

$$v = u_1 + u_2 + \cdots + u_k \quad \text{with} \quad u_i \in U_i.$$

Similarly, we say that $U_1, \dots, U_k$ are *independent* if for any $u_1 \in U_1, \dots, u_k \in U_k$,

$$u_1 + \dots + u_k = 0 \implies u_1 = \dots = u_k = 0.$$

Note that this is a direct generalization of the notions we've introduced for vectors: if $v_1, \dots, v_k \in V$ are any vectors and we take $U_i = \langle v_i \rangle$ to be the one-dimensional space spanned by v_i , then to say that the U_i are independent (resp., span V) is the same as saying the v_i are independent (resp., span V).

If a collection $U_1, \dots, U_k \subset V$ of subspaces of V both spans and is independent, that means that *every element* $v \in V$ *is uniquely expressible as a sum* $v = u_1 + \dots + u_k$ *with* $u_i \in U_i$; this gives an isomorphism $V \cong \prod U_i$, and in this situation we say that $V = \oplus U_i$ is the *direct sum* of the subspaces U_i .

Finally, note that if $V = \oplus U_i$ and we choose a basis for each U_i , the union of these bases will be a basis for V ; in particular, we see that

$$\dim V = \sum \dim U_i.$$

4.2 Operators

In the last few lectures, we introduced the notion of a linear map $\phi : V \rightarrow W$ between two vector spaces V and W , and showed that any such map has a particularly simple form: if we choose bases v_i and w_j for V and W appropriately, the map ϕ is given simply by

$$\phi(v_i) = \begin{cases} w_i, & \text{if } i \leq r; \text{ and} \\ 0, & \text{if } i > r. \end{cases}$$

Now, though, we're looking at a map $\phi : V \rightarrow V$ from a vector space V to itself, and the rules of the game have changed: instead of being free to choose bases for V and W independently, we require that we use the same basis for both the source and target. To put it another way: in the last lecture we classified $n \times m$ matrices A up to the equivalence relation

$$A \sim Q^{-1}AP$$

for any invertible $n \times n$ matrices P and $m \times m$ matrices Q ; we saw that every $n \times m$ matrix A is equivalent to a matrix of the form

$$\begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}.$$

Now, by contrast, our goal is to classify $n \times n$ matrices A up to the equivalence relation

$$A \sim P^{-1}AP$$

for any invertible $n \times n$ matrix P ; and this is a much more difficult task.

Another way in which the theory of operators is fundamentally different from the theory of linear maps is that *we can compose operators* on a given space V : given two maps ϕ and $\psi : V \rightarrow V$, we can compose them to arrive at a map $\phi \circ \psi : V \rightarrow V$. The significance of this is that, while $\text{Hom}(V, W)$ is just a vector space over k , the space $\text{Hom}(V, V)$ has in addition the structure of a (noncommutative) ring. Thus, for example, if we're given an operator $\phi : V \rightarrow V$, we can ask about powers $\phi^m : V \rightarrow V$ and how they behave.

Of course, we commemorate this shift with (how else?) a flurry of new language and notation. First, of course, a linear map from a vector space to itself is called an *operator*, or *endomorphism*, rather than just a linear map, and is typically denoted by a capital Roman letter like T rather than a Greek letter like ϕ . Moreover, the space $\text{Hom}(V, V)$ of all such maps is denoted $\text{End}(V)$, or (in Axler) $\mathcal{L}(V)$.

4.2.1 Our goal

Our goal in studying operators is much the same as in any mathematical theory: we want to take a given object and express it in terms of simpler ones. What does this mean in our present setting? Well, one example might be: if $T : V \rightarrow V$ is any operator, we might try to find subspaces $U_1, U_2 \subset V$ such that $T(U_i) \subset U_i$, and such that $V = U_1 \oplus U_2$. In that case, we have two operators $T_i : U_i \rightarrow U_i$ on smaller-dimensional spaces U_i , and T is just the direct sum of these. Another way to say this: if we choose a basis $v_1, \dots, v_n$ for V consisting of the union of a basis $v_1, \dots, v_k$ for U_1 and a basis $v_{k+1}, \dots, v_n$ for U_2 , then in terms of this basis the matrix representative A of

T will be *block diagonal*; it'll look like

$$A = \begin{pmatrix} A_1 & 0 \\ 0 & A_2 \end{pmatrix}$$

where A_i is the matrix representative of $T_i := T|_{U_i}$. We can then try to break down the operators T_i similarly.

What would be an extreme example of success in this undertaking? That's simple to say: if we can write V as a direct sum $\oplus U_i$ with all the U_i one-dimensional, and such that $T(U_i) \subset U_i$. A linear map on a one-dimensional vector space is necessarily just multiplication by a scalar, so if $v_i \in U_i$ is any nonzero vector we have

$$T(v_i) = \lambda_i v_i$$

for some scalar $\lambda_i \in k$. In this setting, the vectors $v_1, \dots, v_n$ will form a basis for V , and in terms of this basis the matrix representative of T will be *diagonal*: it'll look like

$$A = \begin{pmatrix} \lambda_1 & 0 & 0 & \dots & 0 \\ 0 & \lambda_2 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \dots & \vdots \\ 0 & 0 & 0 & \dots & \lambda_n \end{pmatrix}$$

Such an operator is relatively simple to visualize: it simply expands or contracts along each axis.

Time for some language! First of all, if $T : V \rightarrow V$ is any operator, an *eigenvector* for T is a nonzero vector v such that $T(v) = \lambda \cdot v$ for some scalar $\lambda \in k$; the scalar λ is called the *eigenvalue* of v . If we can find a basis for V consisting of eigenvectors of T , then in terms of this basis the matrix representative of T will be diagonal; in this case we say that T is *diagonalizable*.

All of the above leads us to the obvious question: is every operator diagonalizable? The answer is “no,” but we can come pretty close; the next lecture or two will be devoted to saying what we can do in general.

4.2.2 Two examples

Before we go on, here are two examples of operators that are not diagonalizable.

Example 4.2.1 Consider the operator $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by the matrix

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix};$$

in other words, T sends the first basis vector e_1 to itself, and sends the second basis vector e_2 to $e_1 + e_2$. This is easy to visualize: T fixes the x -axis $\langle e_1 \rangle$, and sends each horizontal line to itself by a shift depending on the height of the line. (This is sometimes called a *shear*.) The first basis vector e_1 is an eigenvector for T with eigenvalue 1; but it's not hard to see from the picture that there are no eigenvectors for T other than multiples of e_1 , so T is not diagonalizable.

Note: we chose to work over $\mathbb{R}$ in this example for the sake of visualization, but it's not hard to see that the same conclusion holds if we replace $\mathbb{R}$ by any field.

Example 4.2.2 Consider the operator $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by the matrix

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix};$$

or more generally,

$$A = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

for any angle θ . These are just rotations around the origin in $\mathbb{R}^2$, and so it's clear that T has no eigenvectors at all.

By contrast with the first example, the conclusion here is very much dependent on the choice of field. For example, if we replace $\mathbb{R}^2$ by $\mathbb{C}^2$, then the operator T has eigenvectors, and indeed is diagonalizable: for example, if we take $v = (1, i)$, then

$$Tv = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ i \end{pmatrix} = \begin{pmatrix} -i \\ 1 \end{pmatrix} = -i \cdot v$$

so v is an eigenvector for T with eigenvalue $-i$.

Exercise 4.2.3 Find another eigenvector for T , and show that T may be diagonalized over $\mathbb{C}$.

Chapter 5

Upper triangular

5.1 Two facts about eigenvectors

5.1.1 Eigenvectors are independent

Proposition 5.1.1 *Let $T : V \rightarrow V$ be any operator. If $v_1, \dots, v_k \in V$ are eigenvectors for T with distinct eigenvalues $\lambda_1, \dots, \lambda_k$, then $v_1, \dots, v_k$ are linearly independent.*

Proof. Suppose that $v_1, \dots, v_k$ are linearly dependent. First of all, we can assume that $v_1, \dots, v_k$ are a minimal linearly dependent set; that is, no proper subset of $\{v_i\}$ is linearly dependent. Now suppose we have a relation of linear dependence among the v_i ; that is, we have

$$c_1 v_1 + \dots + c_k v_k = 0$$

with not all $c_i = 0$; by the hypothesis of minimality, none of the coefficients c_i can be 0.

Now apply T to the linear combination $c_1 v_1 + \dots + c_k v_k$; we have

$$0 = T(c_1 v_1 + \dots + c_k v_k) = \lambda_1 c_1 v_1 + \dots + \lambda_k c_k v_k.$$

Next, multiply the first relation above by λ_k and subtract from the second; we arrive at

$$(\lambda_1 - \lambda_k) c_1 v_1 + \dots + (\lambda_{k-1} - \lambda_k) c_{k-1} v_{k-1} = 0.$$

Since the λ_i are all distinct, this is a nontrivial linear among $v_1, \dots, v_{k-1}$, contradicting the hypothesis that $\{v_1, \dots, v_k\}$ was a minimal linearly dependent collection.

5.1.2 Every operator over $\mathbb{C}$ has an eigenvector

Proposition 5.1.1 holds for operators over any field whatsoever. By contrast, the next statement holds only under the hypothesis that our field is algebraically closed.

Proposition 5.1.2 *If $T : V \rightarrow V$ is any operator on a finite-dimensional vector space over an algebraically closed field k , then T has an eigenvector.*

Proof. This is a good introduction to how we can use the ring structure on $\text{Hom}(V, V) = \text{End}(V)$. Consider all powers of T in $\text{End}(V)$, that is,

$$1, T, T^2, T^3, \dots \in \text{End}(V).$$

(Since $\text{End}(V)$ is a ring, we are writing the identity as 1.) As we've seen, $\text{Hom}(V, V)$ is a finite-dimensional vector space, and so there must exist a linear relation among these; that is, for some $a_0, a_1, \dots, a_m \in k$ we have

$$a_m T^m + a_{m-1} T^{m-1} + \dots + a_1 T + a_0 = 0$$

as linear maps $V \rightarrow V$. But as we've seen, *every polynomial over an algebraically closed field factors completely*; thus we can write

$$a_m x^m + a_{m-1} x^{m-1} + \dots + a_1 x + a_0 = a_m (x - \mu_1)(x - \mu_2) \cdots (x - \mu_m),$$

and correspondingly we have

$$a_m (T - \mu_1)(T - \mu_2) \cdots (T - \mu_m) = 0$$

as maps $V \rightarrow V$. Now we make one basic observation: the composition of isomorphisms is an isomorphism. Correspondingly, we can deduce from the last relation that *for at least one value of i , the linear map $T - \mu_i : V \rightarrow V$ must have a kernel*. In other words, there must be a nonzero vector $v \in V$ such that $(T - \mu_i)v = 0$, and there's our eigenvector.

It's instructive to see how this plays out in the case of Example 4.2.2 of the last lecture. There, we looked at the operator $A : k^2 \rightarrow k^2$ given by the matrix

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Squaring the matrix A , we see that

$$A^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Thus the linear map A satisfies the polynomial relation

$$A^2 + 1 = 0.$$

Over the field $k = \mathbb{R}$, this polynomial has no roots; but over $\mathbb{C}$ it factors: we have

$$A^2 + 1 = (A + i)(A - i)$$

and we can correspondingly find the eigenvectors for A by looking at the kernels of $A + i$ and $A - i$.

5.2 Upper triangular form

Proposition 5.1.2 assures us that over an algebraically closed field every endomorphism of a finite-dimensional vector space has an eigenvector. That's a long way from having a basis of eigenvectors, but we can get a fair amount out of the proposition by applying it iteratively. To set this up, one more bit of language:

Definition 5.2.1 Let V be an n -dimensional vector space. By a *flag* in V , we will mean a nested sequence of linear subspaces

$$0 = V_0 \subset V_1 \subset \cdots \subset V_{n-1} \subset V_n = V$$

where $\dim(V_i) = i$.

In these terms, we can state our main result:

Proposition 5.2.2 *If $T : V \rightarrow V$ is any operator on an n -dimensional vector space over an algebraically closed field k , then there exists a flag $V_1 \subset \cdots \subset V_{n-1} \subset V$ preserved by T ; that is, such that*

$$T(V_i) \subset V_i \quad \forall i.$$

Before we prove this, let's state it in a more concrete (but equivalent) form:

Proposition 5.2.3 *If $T : V \rightarrow V$ is any operator on an n -dimensional vector space over an algebraically closed field k , then there exists a basis $v_1, \dots, v_n$ for V such that the matrix A representing T in terms of this basis is upper triangular; that is, $a_{i,j} = 0$ for $j < i$.*

Proof of the equivalence of 5.2.2 and 5.2.3. The point is, to say that the matrix A representing T in terms of a basis $\{v_i\}$ is upper triangular is simply to say that for each i , $T(v_i)$ is a linear combination of $v_1, v_2, \dots, v_i$. So in one direction: given a flag $V_1 \subset \dots \subset V_{n-1} \subset V$ preserved by T , we let $v_1 \in V_1$ be any nonzero vector; we expand $\{v_1\}$ to a basis for V_2 by adding a vector $v_2 \in V_2 \setminus V_1$; we expand $\{v_1, v_2\}$ to a basis for V_3 by adding a vector $v_3 \in V_3 \setminus V_2$, and so on; in terms of this basis the matrix representative of T will be upper triangular. The other direction is similar: given a basis in terms of which the matrix representing T is upper triangular, we define a flag by $V_i := \langle v_1, v_2, \dots, v_i \rangle$.

Now to prove Proposition 5.2.2, we're going to use quotient spaces. If you're allergic to quotient spaces you can do it without them, and I'd urge you to write out a proof without them; but this is a good illustration of how quotient spaces arise and can be used. In any event, I want to state a couple basic facts about them.

Lemma 5.2.4 *Let V be any vector space, $U \subset V$ a subspace, V/U the quotient space and $\pi : V \rightarrow V/U$ the quotient map. We have natural bijections*

1. $\{\text{subspaces of } V/U\} \longleftrightarrow \{\text{subspaces of } V \text{ containing } U\}$; and
2. $\{\text{maps } \phi : V/U \rightarrow W\} \longleftrightarrow \{\text{maps } \psi : V \rightarrow W \text{ with } U \subset \ker(\psi)\}$.

These are pretty straightforward analogues of the corresponding statements for quotient groups: for the first, the map is simply taking a subspace $W \subset V/U$ to $\pi^{-1}(W)$; for the second, we associate to any $\phi : V/U \rightarrow W$ the composition $\phi \circ \pi$. I'd urge you to write out arguments for them, and as always, if you don't see what to do, ask!

Proof of Proposition 5.2.2. After all that, this is pretty simple. To begin with, we know that T has an eigenvector v ; we'll take the first element of our flag to be simply $V_1 := \langle v \rangle$. Now let U be the quotient space $U = V/V_1$, with $\pi : V \rightarrow U$ the quotient map. We know that $T(V_1) \subset V_1$, so $V_1 \subset \ker(\pi \circ T)$ and by the lemma we have an induced map $S : U \rightarrow U$ making the diagram

$$\begin{array}{ccc} V & \xrightarrow{T} & V \\ \pi \downarrow & & \downarrow \pi \\ U & \xrightarrow{S} & U \end{array}$$

commute. Now we apply Proposition 5.1.2 again, and see that S has an eigenvector $u \in U$; if we take our second flag element V_2 to be $V_2 := \pi^{-1}\langle u \rangle$ we see that $T(V_2) \subset V_2$ as well, and we keep going in this way until we arrive at a full flag.

5.2.1 Properties of upper-triangular operators

While we're still far from our goal, the fact that an arbitrary operator (over an algebraically closed field!) can be represented by an upper triangular matrix has consequences that will get us closer. To state these, let $T : V \rightarrow V$ be any operator that is represented in terms of a basis $v_1, \dots, v_n$ for V by an upper-triangular matrix A , and let $V_k := \langle v_1, \dots, v_k \rangle$ be the corresponding flag.

Proposition 5.2.5 *T is an isomorphism if and only if all diagonal entries of A are nonzero.*

Proof. First, suppose that the diagonal entry $a_{k,k} = 0$. Then $T(V_k) \subset V_{k-1}$, so T must have a kernel and cannot be an isomorphism. In the other direction, suppose v is any nonzero vector such that $T(v) = 0$. We have $v \in V_k \setminus V_{k-1}$ for some k ; that is, we can write

$$v = c_1 v_1 + \dots + c_{k-1} v_{k-1} + c_k v_k \quad \text{with} \quad c_k \neq 0$$

It follows that we can write

$$T(v) = c'_1 v_1 + \dots + c'_{k-1} v_{k-1} + a_k c_k v_k$$

for some $c'_1, \dots, c'_{k-1}$; since $a_{k,k}c_k \neq 0$, $T(v)$ can't be 0.

As an immediate consequence of this, we have the

Corollary 5.2.6 *If $T : V \rightarrow V$ is an operator represented in terms of a basis $v_1, \dots, v_n$ for V by an upper-triangular matrix A , then the eigenvalues of T are exactly the diagonal entries of A . In particular, if the diagonal entries of A are distinct, then T is diagonalizable.*

Proof. This is sort of immediate: λ is an eigenvalue of T if and only if $\ker(T - \lambda) \neq 0$ if and only if λ is equal to one of the diagonal entries of A .

So, it would seem that “most of the time” we succeed: we find an upper-triangular matrix A representing T , and as long as the diagonal entries of A are distinct it turns out T is diagonalizable. But we still have to deal with the case when there are repeated entries in the diagonal of A , and that's what will occupy us for the remainder of this lecture and the next.

The problem is exemplified by the operator T on k^2 given by the matrix $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$; that is, the map sending e_1 to e_1 and e_2 to $e_1 + e_2$. The scalar 1 occupies both diagonal entries in A , but it contributes only one eigenvector; and the corollary above says there can't be any other eigenvectors. So this operator is *not* diagonalizable no matter what field k is; we'd like to understand why not and see if there's a “next best” alternative.

As always, some language is useful. First of all, we'd like to define the *multiplicity* of a given eigenvalue λ of an operator $T : V \rightarrow V$ as the number of times λ appears on the diagonal of an upper-triangular matrix representing T (so that for example we'd say that the operator T above has eigenvalue 1 “with multiplicity 2”). The problem is, that's not a good definition: a priori, it could depend on the choice of upper-triangular matrix representative. So the first thing we have to do is give a more intrinsic characterization of this number. We'll do this in the next lecture, and use it to finish our analysis of operators.

Chapter 6

Jordan canonical form

Today I want to wrap up the discussion of operators on finite-dimensional vector spaces. Throughout this discussion, we will be working with operators on vector spaces over an algebraically closed field K (without the hypothesis of algebraic closure an operator need not have any eigenvectors at all).

To recall where we are, our general goal is to express a given operator $T : V \rightarrow V$ on a vector space V as a direct sum of simpler ones; that is, we find to find subspaces $U, W \subset V$ such that T carries each of U and W to itself, and such that $V = U \oplus W$. In this case, if we took a basis for V consisting of the union of bases for U and W , the matrix representative for T would be block diagonal

$$\begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}$$

In some sense, the ideal outcome in this respect would be if we could break V up into a direct sum of 1-dimensional subspaces V_α , each carried into itself by T . Of course, a linear map on a 1-dimensional vector space is necessarily multiplication by a scalar, so T would act on each V_α by multiplication by a scalar λ_α , and in terms of a basis for V consisting of one nonzero vector from each V_α the matrix representative of T would be the diagonal matrix

$$\begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & \lambda_n \end{pmatrix}$$

Of course, we can achieve this only if V has a basis of eigenvectors for T , and while this may be the case “most” of the time,¹ it’s not always the case. The issue is repeated eigenvalues: as we’ve seen, we can represent T by an upper-triangular matrix A , and the eigenvalues of T are exactly the diagonal entries of A . If those are distinct, then we can find an eigenvector for each, and these will form a basis; but if an eigenvalue is repeated there may still be up to scalars only one eigenvector with that eigenvalue. This is exemplified by our standard example, the linear map represented by the matrix

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

Here the eigenvalue 1 is repeated, but there’s only one eigenvector with that eigenvalue, and the operator is in fact not diagonalizable.

6.1 Generalized kernels and images

The way to resolve this problem is to introduce two new notions, the *generalized kernel* and the *generalized image* of an operator $T : V \rightarrow V$. For the first, consider the nested sequence of subspaces of V :

$$\ker(T) \subset \ker(T^2) \subset \ker(T^3) \subset \cdots \subset V$$

Since these are all subspaces of the finite-dimensional vector space V , they must stabilize; that is, for some m we must have

$$\ker(T^m) = \ker(T^{m+1}) = \ker(T^{m+2}) = \cdots$$

We define the *generalized kernel* of T to be this limit; that is,

$$\text{gker}(T) := \bigcup_m \ker(T^m) = \{v \in V \mid T^m v = 0 \text{ for some } m\}$$

In fact, as soon as two of the spaces $\ker(T^k)$ are equal, all the ones after are equal as well, as Exercise 6.1.1 below asserts. Thus, the generalized

¹I’m sorry to cheese out here, but the correct statement is that in the space $\text{End}(V)$ of linear maps $T : V \rightarrow V$, the subset of those that are diagonalizable contains a dense open subset of $\text{End}(V)$, and “dense” and “open” are words we won’t be introducing until 55b.

kernel of T is equal to $\ker(T^n)$, where n is the dimension of V (which is how it appears in Axler).

Exercise 6.1.1 Let $T : V \rightarrow V$ be an operator on a finite-dimensional vector space V .

1. Show that if $\ker(T^m) = \ker(T^{m+1})$, then $\ker(T^k) = \ker(T^m)$ for all $k \geq m$.
2. Show that if $\operatorname{Im}(T^m) = \operatorname{Im}(T^{m+1})$, then $\operatorname{Im}(T^k) = \operatorname{Im}(T^m)$ for all $k \geq m$.

In similar fashion, we can look at the images of successive powers of T , though in this case we get a decreasing sequence of subspaces

$$V \supset \operatorname{Im}(T) \supset \operatorname{Im}(T^2) \supset \operatorname{Im}(T^3) \supset \dots$$

Again, this has to stabilize, and we define the *generalized image* of T to be this limit; that is,

$$\operatorname{gim}(T) := \bigcap_m \operatorname{Im}(T^m) = \{v \in V \mid v \in \operatorname{Im}(T^m) \text{ for all } m\}$$

As with the generalized kernel, this is equal to $\operatorname{Im}(T^n)$ where $n = \dim V$.

6.1.1 A key fact about generalized kernels and images

Proposition 6.1.2 *If $T : V \rightarrow V$ is any operator on the finite-dimensional vector space V , then*

$$V = \operatorname{gker}(T) \oplus \operatorname{gim}(T)$$

Note that this statement is totally false when applied to the ordinary kernel and image: for example, consider the map $T : k^2 \rightarrow k^2$ given by $T(e_1) = 0$ and $T(e_2) = e_1$. Here the kernel of T is the subspace $\langle e_1 \rangle$ spanned by e_1 , and the image of T is the same subspace!

Proof of Proposition 6.1.2. First, since $\text{gker}(T) = \ker(T^n)$ and $\text{gim}(T) = \text{Im}(T^n)$ for $n = \dim V$, we see that

$$\dim \text{gker}(T) + \dim \text{gim}(T) = n.$$

Thus, to prove that $V = \text{gker}(T) \oplus \text{gim}(T)$ we just have to show that $\text{gker}(T) \cap \text{gim}(T) = 0$. But now observe that since $T(\text{gim}(T)) = \text{gim}(T)$ by definition, any power of T maps $\text{gim}(T)$ to itself isomorphically; thus $\text{gim}(T)$ cannot contain any nonzero element of $\text{gker}(T)$.

6.1.2 Nilpotent operators

A special case of the above is the

Definition 6.1.3 An operator $T : V \rightarrow V$ is called *nilpotent* if $T^m = 0$ for some m ; equivalently, if $\text{gker}(T) = V$ or $\text{gim}(T) = (0)$.

We've seen one example of a nilpotent operator (other than 0) in the example following the statement of Proposition 6.1.2. For a generalization, consider the map $T : k^m \rightarrow k^m$ defined by

$$T(e_1) = 0; T(e_2) = e_1; T(e_3) = e_2 \dots T(e_m) = e_{m-1}.$$

In fact, we have now described *every* nilpotent operator: we have the

Lemma 6.1.4 *Let $T : V \rightarrow V$ be any nilpotent operator. Then we have a direct sum decomposition*

$$V = \bigoplus U_\alpha \quad \text{with} \quad T(U_\alpha) \subset U_\alpha$$

and such that U_α has a basis $v_1, v_2, \dots, v_{m_\alpha}$ with

$$T(v_1) = 0 \quad \text{and} \quad T(v_i) = v_{i-1} \quad \text{for } i = 2, 3, \dots, m_\alpha$$

We'll defer the proof of Lemma 6.1.4 to the end of the lecture. In the meantime, though, here's a restatement of the lemma that avoids the direct sum:

Corollary 6.1.5 *If $T : V \rightarrow V$ is any nilpotent operator, we can find a basis $v_1, v_2, \dots, v_n$ such that for each i , either $T(v_i) = v_{i-1}$ or $T(v_i) = 0$.*

To put this another way, we're saying that a nilpotent operator T can always be represented by a matrix A consisting entirely of 0s except directly above the diagonal; the entries $a_{i,i+1}$ directly above the diagonal will be either 0 or 1.

6.2 Back to eigenvectors and eigenvalues

The notion of generalized kernels is exactly what allows us to get past the problem of repeated eigenvalues. It's very straightforward: just as we define an eigenvector with eigenvalue λ for an operator $T : V \rightarrow V$ to be an element of $\ker(T - \lambda)$, we define a *generalized eigenvector* for T to be an element of $\text{gker}(T - \lambda)$ —in other words, a vector killed by some power of $T - \lambda$.

Here's the point: as we've seen, to say that an operator T is diagonalizable is exactly to say that

$$V = \bigoplus_{\lambda \in k} \ker(T - \lambda).$$

As we've seen, that's not always the case. But here's what is true always:

Proposition 6.2.1 *If T is any operator on the vector space V , we have*

$$V = \bigoplus_{\lambda \in k} \text{gker}(T - \lambda);$$

in other words, V always has a basis of generalized eigenvectors for a given operator T .

Proof. We simply induct on the dimension of V . We do know that T has at least one eigenvector; that is, for some λ we have $\text{gker}(T - \lambda) \neq 0$. Now we invoke Proposition 6.1.2 to write

$$V = \text{gker}(T - \lambda) \oplus \text{gim}(T - \lambda)$$

and apply the inductive hypothesis to $\text{gim}(T - \lambda)$.

The notion of generalized kernel allows us to define the multiplicity of an eigenvalue correctly:

Definition 6.2.2 If $T : V \rightarrow V$ is an operator, we define the *multiplicity* of an eigenvalue λ to be the dimension of the generalized kernel of $T - \lambda$.

Exercise 6.2.3 Let $T : V \rightarrow V$ be an operator and A an upper-triangular matrix representing T . Show that the multiplicity of an eigenvalue λ of T is equal to the number of times λ appears in the diagonal of A . Hint: this boils down to the observation that a strictly upper-triangular matrix A (that is, such that $a_{i,j} = 0$ for $j \leq i$) is nilpotent.

6.3 Jordan canonical form

Let's now put together what we've seen. Suppose we're given an operator T on the vector space V . To start with, we have the direct sum decomposition

$$V = \bigoplus_{\lambda \in k} \text{gker}(T - \lambda).$$

Now let's look at each summand $U_\lambda := \text{gker}(T - \lambda)$. On this subspace, $N := T - \lambda$ is nilpotent, and so by Corollary 6.1.5 we can find a basis for U_λ in terms of which N is represented by a matrix consisting entirely of 0s except for some 1s directly above the diagonal; the operator $T = N + \lambda$ is then represented by a matrix with λ s on the diagonal, 1s and 0s directly above the diagonal and zeroes everywhere else. Putting it all together, we arrive at the

Theorem 6.3.1 (Jordan normal form) *Let $T : V \rightarrow V$ be any operator, with eigenvalues $\lambda_1, \dots, \lambda_m$. Then there is a basis for V in terms of which the matrix of T is block diagonal, with the i th block consisting of λ_i s on the diagonal, 0s and 1s directly above the diagonal, and 0s everywhere else.*

6.3.1 Proof of Proposition 6.1.4

First, let's recall the statement:

Lemma 6.3.2 *Let $T : V \rightarrow V$ be any nilpotent operator. Then we have a direct sum decomposition*

$$V = \bigoplus U_\alpha \quad \text{with} \quad T(U_\alpha) \subset U_\alpha$$

and such that U_α has a basis $v_1, v_2, \dots, v_{m_\alpha}$ with

$$T(v_1) = 0 \quad \text{and} \quad T(v_i) = v_{i-1} \quad \text{for } i = 2, 3, \dots, m_\alpha$$

Proof. Once more, the proof is by induction on $\dim V$. To start, since T is nilpotent, the image $W := \text{Im}(T)$ is a proper subspace of V , so we can apply the inductive hypothesis and find a basis

$$\{v_i^\alpha \mid 1 \leq i \leq m_\alpha\}$$

for $\text{Im}(T)$ with $T(v_1^\alpha) = 0$ and $T(v_i^\alpha) = v_{i-1}^\alpha$ for $i > 1$. Now, all of these are in the image of T , so for each α we can find a vector $v_{m_\alpha+1}^\alpha$ such that $T(v_{m_\alpha+1}^\alpha) = v_{m_\alpha}^\alpha$. We now have a larger collection of vectors

$$\{v_i^\alpha \mid 1 \leq i \leq m_\alpha + 1\},$$

and to finish the argument we want to show that *these vectors are independent, and can be enlarged to a basis for V by adding elements of $\ker(T)$.*

For the first of these, suppose we had a relation of linear dependence

$$\sum_{\alpha} \sum_{i=1}^{m_\alpha+1} c_i^\alpha v_i^\alpha = 0.$$

Applying T to this equality, we kill the terms with $i = 1$ and arrive at

$$\sum_{\alpha} \sum_{i=2}^{m_\alpha+1} c_i^\alpha v_{i-1}^\alpha = 0.$$

Since this is a linear relation among the basis vectors for $\text{Im}(T)$, all the coefficients must be zero; in other words, the only potentially nonzero coefficients

in the original linear relation are the c_1^α . But the vectors v_1^α are part of our original basis, so any linear relation among these must be trivial; thus all $c_i^\alpha = 0$ and we see that our vectors $\{v_i^\alpha \mid 1 \leq i \leq m_\alpha + 1\}$ are indeed independent.

Finally, complete the collection $\{v_i^\alpha \mid 1 \leq i \leq m_\alpha + 1\}$ to a basis for V by adding vectors $w_1, \dots, w_\ell$. For each j , $T(w_j)$ is in the image of T and hence equal to a linear combination

$$T(w_j) = \sum_{\alpha} \sum_{i=1}^{m_{\alpha}} c_i^{\alpha} v_i^{\alpha}$$

of our original basis vectors $\{v_i^\alpha \mid 1 \leq i \leq m_\alpha\}$. If we set

$$u_j := \sum_{\alpha} \sum_{i=1}^{m_{\alpha}} c_i^{\alpha} v_{i+1}^{\alpha}$$

then since u_j is a linear combination of the vectors $\{v_i^\alpha \mid 1 \leq i \leq m_\alpha + 1\}$ we can replace w_j by $w_j - u_j$, and this will still be a basis for V ; plus, $w_j - u_j \in \ker(T)$, as desired.

Chapter 7

Moving on

7.1 Characteristic and minimal polynomials

I want to clean up one loose end from our discussion of operators. If you recall, when we first started analyzing an operator $T : V \rightarrow V$ on a vector space V of dimension n , a key idea was the observation that since $\dim(\text{End}(V)) = n^2$, the elements

$$1, T, T^2, T^3, \dots, T^{n^2} \in \text{End}(V)$$

could not all be linearly independent. In other words, there had to be a polynomial $P(x) \in k[x]$, of degree at most n^2 , such that $P(T) = 0 \in \text{End}(V)$; we used this observation to say that any operator on a finite-dimensional vector space over an algebraically closed field had an eigenvector. The degree of the polynomial P did not enter into our analysis, but I want to take up the question now, simply because we can do better than $\deg(P) \leq n^2$; in fact, we have the

Proposition 7.1.1 *If $T : V \rightarrow V$ is an operator on the n -dimensional vector space V , then T satisfies a polynomial relation $P(T) = 0$ of degree n or less.*

Proof. From our analysis, we saw that we can decompose V as a direct sum of generalized eigenspaces for T ; we have

$$V = \bigoplus_{\lambda \in k} \text{gker}(T - \lambda).$$

For each λ , let $n_\lambda = \dim \text{gker}(T - \lambda)$, so that $\sum n_\lambda = n$. Now $T - \lambda$ is nilpotent on $\text{gker}(T - \lambda)$, so $(T - \lambda)^{n_\lambda} = 0$ on $\text{gker}(T - \lambda)$, and it follows that

$$P(T) := \prod_{\lambda} (T - \lambda)^{n_\lambda} = 0;$$

Thus we have a polynomial of degree n satisfied by T . The polynomial P is called the *characteristic polynomial* of T .

In fact, we can often do better. If we set

$$m_\lambda := \min \{m : (T - \lambda)^m = 0 \text{ on } \text{gker}(T - \lambda)\}$$

then m_λ can be any integer between 1 and n_λ (for example, it's 1 if the Jordan block corresponding to λ is purely diagonal; it's equal to n_λ if the Jordan block has 1s in every entry directly above the diagonal). Thus T satisfies the polynomial

$$Q(T) := \prod_{\lambda} (T - \lambda)^{m_\lambda} = 0;$$

this is the smallest degree polynomial satisfied by T (check this!), and is called the *minimal polynomial* of T .